

AMAN SAMANI

Full Stack Software Engineer

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PROFESSIONAL SUMMARY

Full Stack Software Engineer (B.Tech CS, 2026) specializing in scalable web and mobile applications across the MERN and Next.js stacks. Skilled in production-grade REST API design, AI-integrated features, and secure authentication systems. Shipped HireKarlo, a live AI-powered applicant tracking system, and Fitlip, an AI health app with an adaptive recommendation engine. Strong focus on backend architecture and systems that hold up under real usage.

TECHNICAL SKILLS

Languages	JavaScript (ES6+), TypeScript, Python, Java
Core CS	Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks
Frontend	React.js, Next.js, React Native (Expo), HTML5, CSS3
Backend	Node.js, Express.js, REST API Design, JWT Authentication, Middleware
Databases	MongoDB, PostgreSQL, SQL, Redis
AI / ML	Gemini API, Ollama (local LLM), Scikit-learn, XGBoost, Pandas, Vector Search (ChromaDB)
Tools & DevOps	Git, GitHub, Prisma, Postman, Vercel, Railway

PROJECTS

AI Personal Health Assistant — Fitlip

React Native · Node.js · MongoDB · Redis · Gemini API

- Building a cross-platform mobile app delivering AI-personalized nutrition and workout plans (in active development); backend ran live on Railway for 2 months during testing, now developed locally while adding features.
- Designed an adaptive calorie-adjustment engine that evaluates 7-day adherence and weight-change trends to auto-tune targets weekly, on top of a Mifflin-St Jeor BMR/TDEE and macro-split calculation core.
- Integrated Google's Gemini API for AI meal-plan generation with Zod schema validation and a **post-generation correction algorithm** that clamps AI output to calorie budgets, preventing overshoot from LLM inconsistency.
- Built a background job pipeline with BullMQ/Redis (retry + exponential backoff) to run scheduled weekly adjustments and trigger Expo push notifications.
- Designed **42+ REST API endpoints** across a full auth system (JWT, bcrypt, Google OAuth, OTP recovery) plus nutrition, workout, and tracking modules, backed by a curated 300+ item Indian/international food database.

HireKarlo — AI Applicant Tracking System · [Live Demo](#)

Next.js · TypeScript · PostgreSQL · Prisma · Gemini API

- Built and deployed a full-stack ATS where recruiters post jobs and candidates apply through a public, no-login link; live in production.
- Integrated Gemini to parse uploaded resumes (PDF/DOCX) server-side and score each candidate **0-100** against the job description, persisting results instantly to the pipeline view; scoring fails open so a flaky AI call never blocks a submission.
- Enforced authorization on every server action by re-verifying job ownership server-side rather than trusting a client-side role check, keeping each recruiter's data fully isolated.
- Built a Kanban-style pipeline (Applied → Screening → Technical → HR → Offer/Rejected → Hired) with interview scheduling, an append-only activity log, and automatic email notifications on every stage change.

Fraud Detection — Transaction Classification Model

Python · XGBoost · scikit-learn

- Built an end-to-end ML pipeline to classify fraudulent transactions in a severely imbalanced dataset.
- Applied SMOTE oversampling to address class imbalance, then trained an XGBoost classifier (300 estimators) on the resampled data.
- Tuned the decision threshold to optimize F1-score, achieving **91.1% recall and 80.5% F1-score** (72.1% precision) on a held-out test set of 465K+ transactions.
- Achieved a **0.995 ROC-AUC**, correctly identifying 479 of 526 fraudulent transactions in testing.

Rookie Agent — Local AI Dev-Memory System *(in progress)*

Python · Ollama · ChromaDB · NetworkX · SQLite

- Building a local-first system that turns git history into structured memory: commit diffs run through a local Llama 3.2 model (Ollama) to extract project/skill/bug metadata, persisted to SQLite and a NetworkX knowledge graph, with duplicate-commit detection to skip reprocessing.
- Building a semantic search layer on ChromaDB with a custom Hinglish-tuned embedding model, alongside a privacy-first activity watcher that regex-scrubs emails, API keys, and other secrets before anything is stored.
- Next up: an MCP server exposing memory as tools for Claude Desktop, and a confirm-before-execute action gate for agent-initiated tasks.

LEADERSHIP & ACHIEVEMENTS

- Led a 4-member cross-functional team (backend, frontend, UI/UX, research) as Project Lead for a final-year capstone, presented to faculty advisor Dr. Harsh Dev.
- Active problem solver on LeetCode in Java, with focus on arrays, strings, and dynamic programming.

EDUCATION

Babu Banarsi Das University, Lucknow
B.Tech in Computer Science

Sept 2022 – June 2026
GPA: 7.6 / 10.0